

ABSTRACT

Increased consumption and demand for freshwater has resulted to massive production of wastewater, which has toxins and is harmful to the environment if not treated. The use of **permeable reactive barriers** using activated carbon has proven to be a low cost and environmental friendlier solution to the problem compared to conventional methods. This minireview demonstrated the use of bamboo activated carbon, which is mesoporous in treating wastewater pollutants through adsorptive processes. The paper discusses the chemical and physical processes involved in activation and **carbonization** of bamboo charcoal, its adsorptive and structural characteristics. Case examples of its application in **wastewater treatment** based on literature review are also evaluated. Overall, bamboo is a viable renewable biomass material used in **wastewater treatment** compared to alternative carbon-based materials such as wood charcoal, coconut shell activated charcoal and **carbon nanotubes**.

Statement of the problem

Access to clean and safe drinking water is a major challenge in many communities around the camp. Most available water sources contain impurities like dirt, harmful chemicals, and microorganisms that can cause diseases. Drinking contaminated water can lead to health problems such as diarrhea, cholera, and typhoid.

Modern water purification methods are often expensive and not easily available to everyone, especially in rural areas. Therefore, there is a need to find a simple, affordable, and effective method of purifying water using locally available materials.

This project was carried out to investigate whether charcoal can be used to purify dirty water and improve its quality, making it safer for use.

Objectives

- To demonstrate a simple and affordable method of water purification.
- To investigate the effectiveness of charcoal in purifying dirty water.

Background information

Water is one of the most important natural resources for human survival. It is used for drinking, cooking, cleaning, and other daily activities. However, not all water available is clean and safe for human consumption. Many water sources such as rivers, lakes, and wells contain impurities like dirt, chemicals, and microorganisms, which can cause diseases such as cholera, typhoid, and diarrhea.

According to the World Health Organization, access to clean and safe drinking water is essential for maintaining good health and preventing waterborne diseases. Due to lack of proper purification systems, many communities rely on simple and affordable methods to improve water quality.

Charcoal has been used for many years as a filtering material because of its ability to remove impurities, bad odour, and colour from water. Charcoal works through a process called adsorption, where harmful substances stick to its surface as water passes through it. This makes water clearer and safer for use.

WATER PURIFICATION USING CHARCOAL

Apparatus/ materials

Dirty water

500ml

Gravel

400g

Crushed charcoal

250g

Cotton wool

500g

A plastic bottle

5 litres

Fine sand (washed)

300g

Scalpel

1

Procedure

01

Cut the plastic bottle into half horizontally

02

Use the upper part of the bottle as a funnel and the lower part as the beaker

03

Take the cotton wool and place a layer on the funnel, then add crushed charcoal, a layer of fine sand is added on top followed by a layer of gravel.

04

After arranging all the required materials in order, you pour dirty water into the funnel (the funnel should not be over filled)



Variables

Dependent variables

- Color of water
- Cleanliness of water
- Presence of dirty particles

Independent variables

- Type of filter materials used
- Amount of charcoal used
- Particle size of the charcoal (powder vs small chunks)

❑ More charcoal or thicker layer usually provides more surface area, which can adsorb more contaminants.

Smaller charcoal particles may filter better but can slow water flow.

Data

Parameter	Before filtration	After filtration
color	Brown/muddy	clear
volume	500ml	470 ml
Visible particles	present	Not visible
Smell	Unpleasant smell	Smell reduced

Data analysis and interpretation

The results show that charcoal filtration improved the quality of water.

- The colour changed from brown to clear, indicating removal of dirt and impurities.
- The bad smell was removed, showing charcoal absorbed unwanted substances.
- The number of particles reduced, meaning charcoal trapped solid impurities.
- The water became clearer, showing improvement in physical purity.

This happened because charcoal **has** tiny pores that trap impurities, making the water cleaner.

Data Discussion

Before purification, many visible particles were observed. After filtration, the number of particles reduced significantly. This confirms that filter materials acted as physical barriers preventing solid impurities from passing through, the unpleasant smell was reduced as charcoal absorbed organic compounds responsible for bad odour. The water becomes less cloudy after filtration as the filter layers removed the fine particles that causes turbidity.

Conclusion

In conclusion, the project proved that water purification is an important process for improving water quality and making it safer for human use. The use of locally available materials such as charcoal provides a simple, affordable, and effective method of purifying water, especially in areas where clean water is limited.

Overall, the objectives of the project were achieved, and the experiment confirmed that water purification is essential for protecting human health and preventing water-borne diseases.

Recommendations

- Charcoal and sand should be replaced regularly because after some time they become saturated with impurities and become less effective.
- This method should be promoted especially in rural areas where access to clean and safe water is limited.
- More research should be done to improve the purification system and to test other locally available materials that can improve water quality.

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